

Luca Choteborsky

Production Engineer at Meta — M.Eng Computer Science, University of Cambridge
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PROFILE

Production Engineer at Meta; M.Eng Computer Science from the University of Cambridge with coursework in ML Systems (PyTorch) and distributed data processing. Built and operated distributed infrastructure supporting AI training clusters at scale, with hands-on experience in Kubernetes orchestration.

ACADEMICS

University of Cambridge M.Eng Computer Science

October 2024 – June 2025
Distinction

- Principles of Machine Learning Systems: Studied and implemented algorithms and system techniques in PyTorch supporting training and inference of ML models across a spectrum of computing systems.
- Large-scale data processing and optimisation: Hands-on experience with dataflow programming using TensorFlow and cluster computing; academic grounding in next-generation distributed systems.
- Dissertation: built and evaluated a federated Kubernetes scheduler (Go) using online FPCA-based load prediction and distributed gRPC aggregation, outperforming the default kube-scheduler on utilisation balance and job completion time across ML inference workloads.

University of Cambridge BA Computer Science

October 2021 – June 2024
Double 1st Class

- Machine Learning and Real-world Data: practicals in statistical classification (Naive Bayes), sequence modelling (Hidden Markov Models), and social network analysis; experimental methodology including cross-validation, statistical testing, and evaluation design.
- Concurrent and Distributed Systems: concurrency primitives, distributed algorithms, consistency models, and fault-tolerant message-passing systems.
- Advanced Graphics and Image Processing: rendering equation and path tracing, image-based rendering (light fields and Neural Radiance Fields), optimization-based image processing, and colour science; techniques directly underpinning modern computer vision systems.

TECHNICAL SKILLS

Programming Languages	Python, Go, C++, Rust, C, Java, SQL, OCaml, Bash, $\text{\LaTeX}$
ML & Frameworks	PyTorch, TensorFlow, Kubernetes, Docker, gRPC

PROFESSIONAL EXPERIENCE

Meta

United Kingdom

Production Engineer, Network AI PE

September 2025 – Present

- **NVLINK Bug Discovery & Automated Repair:** Root-caused a configuration misapplication affecting >25% of the NVLINK switch fleet (~40,000 devices). Devised an automated mechanism to detect impacted switches and apply targeted repair fleet-wide.
- **Production Service Migration:** Migrated APIs for a network drain service handling >1 million daily requests; implemented A/B testing to guarantee zero user traffic impact during transition.
- **Drain Monitoring & Observability:** Used existing data collection infrastructure to validate drain state across a fleet of 100,000+ switches; performed aggregations on collected data to prevent alert noise from scaling with fleet size, minimising oncall burden.
- **CEL Compiler CLI Tool (Go):** Added CEL compilation to an existing Go CLI tool, parsing CEL configuration files and generating device domain filters. Enabled dev-time testing across all consuming teams, increasing validation coverage without increasing canary reservations.

Zilliqa

London, United Kingdom

Intern, Software Engineering

July – September 2023

- Modified the Ethereum block explorer Otterscan using React and TypeScript; worked with Ethereum Virtual Machine APIs in C++ for cross-chain interoperability.

PROJECTS

Pronto – Federated Kubernetes Scheduler

Go, Kubernetes Scheduling Framework, gRPC, Docker, FPCA

<https://github.com/lucachot/pronto-framework>

- Evaluated against the default kube-scheduler on CPU-intensive and ML inference workloads, demonstrating improved node utilisation balance and reduced job completion times under batch workloads.
- Built a distributed aggregation layer using gRPC streaming where per-node FPCA estimates are continuously merged into a shared global embedding — a form of federated learning applied to infrastructure load prediction.
- Implemented a custom Kubernetes scheduler plugin via the official Scheduling Framework API, routing pods to nodes based on real-time CPU and memory telemetry rather than declared resource requests.

Light Field Viewer

WebGL, GLSL, JavaScript

<https://github.com/lucachot/lfv-webgl>

- Built a WebGL/GLSL shader pipeline rendering interactive light field scenes from multi-image arrays, with real-time aperture control, focal plane adjustment, and XYZ/orbital camera navigation.
- Wrote a Python preprocessing pipeline to parse and reformat raw light field image data (Stanford datasets) into the texture layout consumed by the GPU renderer.